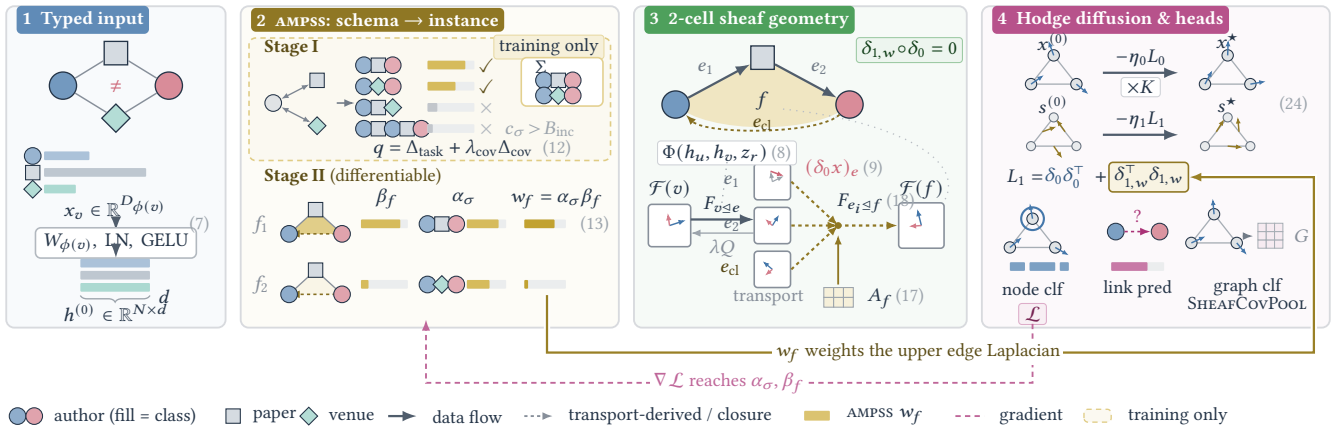

1 Typed input
$x_v \in \mathbb{R}^{D_{\phi(v)}}$
(7)
$W_{\phi(v)}$, LN, GELU
$h^{(0)} \in \mathbb{R}^{N \times d}$
2 AMPSS: schema → instance
Stage I
training only
$c_\sigma > B_{\text{inc}}$
$q = \Delta_{\text{task}} + \lambda_{\text{cov}} \Delta_{\text{cov}}$ (12)
Stage II (differentiable)
β_f
α_σ
$w_f = \alpha_\sigma \beta_f$
f_1
f_2
(13)
3 2-cell sheaf geometry
$\delta_{1,w} \circ \delta_0 = 0$
e_1
e_2
f
e_{cl}
$\Phi(h_u, h_v, z_r)$ (8)
$(\delta_0 x)_e$ (9)
$\mathcal{F}(v)$
$\mathcal{F}(f)$
$F_{v \trianglelefteq e}$
$F_{e_i \trianglelefteq f}$ (10)
λQ
transport
A_f (17)
4 Hodge diffusion & heads
$x^{(0)}$
$x^\star$
$-\eta_0 L_0$
$\times K$
$s^{(0)}$
$s^\star$
$-\eta_1 L_1$
(24)
$L_1 = \delta_0 \delta_0^\top + \delta_{1,w}^\top \delta_{1,w}$
node clf
link pred
graph clf
SHEAFCOVPOOL
G
$\mathcal{L}$
w_f weights the upper edge Laplacian
$\nabla \mathcal{L}$ reaches α_σ, β_f
author (fill = class)
paper
venue
data flow
transport-derived / closure
AMPSS w_f
gradient
training only